





















Named nets in both drawings denote the same interface. Bus labels are conceptual cables, not single copper traces.

Vacuum base with differential drive assumed; NO mecanum wheels. OEM motor control, encoders and safety sensors preferred.

LiDAR stays on OEM wiring provisionally. Access through documented base API or a model-specific adapter to Pi; do not guess pins.

Kinect v1 assumed from user: USB 2.0 + libfreenect. Confirm model label and validate depth streaming on the Pi.

Two separate IMU bus channels avoid equal-address conflicts. Camera IMU must move with camera; publish separate frames.

PCA9685 applies only to PWM servos. Smart bus servos require the matching bus transceiver and feedback protocol.

ESP32 shown USB-powered only; do not also inject external 5V without verified power selection/backfeed protection.

T12 also has an OEM gyro. Body IMU + head IMU remain explicit; OEM gyro access and LiDAR protocol are unverified.

Symbol terminal numbers are logical IDs, NOT connector pin numbers
MODULE INTERCONNECTIONS ONLY — NOT FOR PCB FABRICATION

Sheet: /
File: underbot_connections.kicad_sch

Title: UnderBot system connections — vacuum base, depth, dual IMU

Size: A2	Date: 2026-09-29	**Rev: 0.1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1